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Analytical Results of the Tank 51H Sludge Batch 11 As-Received Qualification Sample

E.J. Craig

October 2024

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REVIEWS AND APPROVALS

AUTHORS:

ELIZABETH CRAIG (Affiliate) Digitally signed by ELIZABETH CRAIG (Affiliate) Date: 2024.11.04 09:42:15 -05'00'

E. J. Craig, Chemical Flowsheet Development Date

TECHNICAL REVIEW:

MATTHEW SIEGFRIED (Affiliate) Digitally signed by MATTHEW SIEGFRIED (Affiliate) Date: 2024.11.04 09:46:34 -05'00'

M. J. Siegfried, Chemical Flowsheet Development, Reviewed per E7 2.60 Date

APPROVAL:

BRANDI CLARK (Affiliate) Digitally signed by BRANDI CLARK (Affiliate) Date: 2024.11.04 09:50:00 -05'00'

B. N. Clark, Chemical Flowsheet Development Date

FRANK PENNEBAKER (Affiliate) Digitally signed by FRANK PENNEBAKER (Affiliate) Date: 2024.11.04 09:52:45 -05'00'

F. M. Pennebaker, Director, Nuclear and Chemical Processing Date

Robert Hoeppel Digitally signed by Robert Hoeppel Date: 2024.11.05 16:37:24 -05'00'

R. M. Hoeppel, DWPF Facility Engineering Manager Date

EXECUTIVE SUMMARY

Savannah River National Laboratory (SRNL) has been requested by Savannah River Mission Completion (SRMC) to qualify the next sludge batch, Sludge Batch 11 (SB11), for processing at the Defense Waste Processing Facility (DWPF). This report documents the first steps of the qualification process, characterization of the Tank 51H as-received qualification material. SRMC sent SRNL two 200 mL samples received from Tank 51H on the 12th of August 2024. The combined sample, HTF-51-24-85/86, was analyzed for the following: supernate and slurry density, weight percent solids, settling, chemical composition, and radionuclides.

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LIST OF ABBREVIATIONS

AR	Aqua Regia
ATFC	Analytical and Tank Farm Characterization
CPC	Chemical Process Cell
DMA	Direct Mercury Analyzer
DWPF	Defense Waste Processing Facility
HDPE	High Density Polyethylene
IC	Ion Chromatography
ICP-ES	Inductively Coupled Plasma-Emission Spectroscopy
ICP-MS	Inductively Coupled Plasma-Mass Spectroscopy
NA	Not Applicable
PF	Peroxide Fusion
RSD	Relative Standard Deviation
SB11	Sludge Batch 11
SRMC	Savannah River Mission Completion
SRNL	Savannah River National Laboratory
TIC/TOC	Total Inorganic Carbon/Total Organic Carbon
TTQAP	Task Technical and Quality Assurance Plan
TTR	Technical Task Request
Wt%	Weight %

1.0 Introduction

Savannah River National Laboratory (SRNL) has been requested to qualify the next sludge batch, Sludge Batch 11 (SB11), for processing at the Defense Waste Processing Facility (DWPF). To accomplish this task, Savannah River Mission Completion (SRMC) sent SRNL two unwashed 200 mL samples received from Tank 51H on the 12th of August 2024 and later sent two 3 L samples to be characterized, washed, and used in DWPF simulations. This report documents the first steps of the qualification process, characterization of the as-received Tank 51H qualification material. These results will be used to support a reprojection of SB11 by SRMC for glass work and Chemical Process Cell (CPC) activities. This report contains analytical results for the as-received qualification sample as requested in the technical task request (TTR).[1] This task is governed by a Task Technical and Quality Assurance Plan (TTQAP).[2]

2.0 Experimental Procedure

2.1 Sample Information

SRMC sent SRNL two 200 mL samples, HTF-51-24-85, and HTF-51-24-86, which were combined into HTF-51-24-85/86 into a 500-mL High Density Polyethylene (HDPE) bottle in the SRNL shielded cells, shown in Figure 2-1. Each doorstop was rinsed with the sludge's supernate to ensure all solids were transferred to the combined sample. From the combined sample, a 50 mL aliquot was subsampled for corrosion control and evaporator feed qualification analyses. The remaining sample was prepared for analysis in the shielded cells via Aqueous Supernate Dilution, Water Diluted Slurry, Peroxide Fusion (PF), and Aqua Regia (AR) methods. Table 2-1 shows the preparation methods used for each analysis performed by the Analytical and Tank Farm Characterization (ATFC) group at SRNL. Weight % solids (wt%), density, and a settling test were performed in the shielded cells.

Table 2-1 Sample Preparations Performed for Each Analysis

Preparation	Slurry/Filtrate	Analysis
Aqueous Supernate Dilution	Filtrate	TIC/TOC, T Base/OH/Other Base Exc CO_3^{2-}, IC Anions, ICP-ES, ICP-MS, DMA Total Mercury, Gamma Scan, Pu238/241, Pu242/Pu244 and U233/U234/U235/U236
Water Diluted Slurry	Slurry	IC Anions (Oxalate)
Aqua Regia	Slurry	ICP-ES, ICP-MS, DMA Total Mercury
Peroxide Fusion	Slurry	ICP-ES, Pu238/241, Pu242/Pu244 and U233/U234/U235/U236

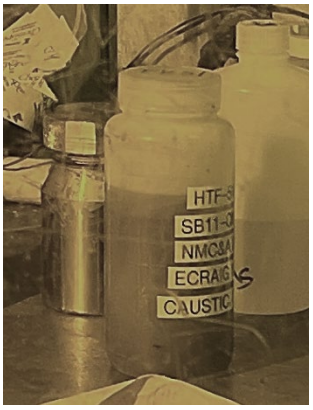


Figure 2-1 The Combined Sample HTF-51-24-85/86 in the Shielded Cells

2.2 Sample Analysis

Slurry densities and supernate densities were obtained in triplicate using 10 mL volumetric flasks and an analytical balance. The supernate was obtained by filtering a portion of the slurry through a 0.45 µm nylon filter. Filtrate, instead of supernate, was used to ensure no insoluble solids would be accounted for in the dissolved solids measurements.

Weight percent total and dissolved solids measurements were performed in quadruplicate. Aliquots of slurry and filtrate were dried to a constant weight at 115 °C.[3] Filtrate was obtained by filtering a portion of the slurry through a 0.45 µm nylon filter. Filtrate, instead of supernate, was used to ensure no insoluble solids would be accounted for in the dissolved solids measurements. Weight percent insoluble and soluble solids were calculated from the total and dissolved solids measurement. To measure calcined solids, the sample preparations used for total solids were further placed into a furnace and heated at 1100 °C for 2 hours.

A settling test was performed on the HTF-51-24-85/86 combined sample. This was performed in a 100 mL graduated cylinder, with a diameter of 2.54 cm, using approximately 100 mL of slurry. Before adding the slurry to the graduated cylinder, HTF-51-23-85/86 was vigorously shaken to ensure the sample was well mixed. The slurry and supernate levels were recorded at various times during the settling process.

Aqueous Supernate Dilutions were performed to obtain supernate analytical results and submitted to SRNL-ATFC for total inorganic carbon/total organic carbon (TIC/TOC), Total Base/OH/Other Base excluding CO_3^{2-}, ion chromatography (IC) anions, inductively-coupled plasma-emission spectroscopy (ICP-ES), inductively-coupled plasma-mass spectroscopy (ICP-MS), direct mercury analyzer (DMA), gamma scan, and alpha counting. Isotopes of uranium were determined from purified aliquots analyzed by ICP-MS through the U233/U234/U235/U236 method. Aliquots of the dissolution were purified with diamyl, amylphosphonate based solid phase extraction.[4] Pu-238 and Pu-241 concentrations were determined via PF by isolating the Pu by a 2-thenoyltrifluoroacetone extraction procedure and analyzed by beta and alpha counting to determine the ratio of beta activity from Pu-241 to the alpha activity from the other isotopes of Pu (Pu-238, Pu-239, Pu-240, and Pu-242).The total alpha activity from the Pu isotopes was determined by alpha counting and ICP-MS.[4] The supernate was obtained by filtering a portion of the slurry through a 0.45 µm nylon filter. Filtrate, instead of supernate, was used for the supernate analytical results to ensure no insoluble solids would be accounted for in the dissolved solids measurements. One set of supernate aliquots were diluted in quadruplicate by ~10x with water. The purpose of dilution was to reduce personnel radiation exposure.

Water diluted slurry samples were performed to obtain oxalate in slurry. This was performed by weighing out a set amount of slurry, diluting the sample with water by ~10x to dissolve oxalate, and filtering the sample with a 0.45 µm Nylon syringe filter to prevent solids from being sent to the IC-anions instrument.

Slurry samples were digested by two methods, AR and PF, for analysis of the slurry solids. For the AR digestions, aliquots of slurry were mixed with aqua regia and heated for two hours in sealed Teflon containers in an oven at 115 °C. The resulting liquids were diluted to 100 mL and submitted to SRNL-ATFC for ICP-ES, ICP-MS, and DMA analyses. For the PF digestions, aliquots of slurry were dried and fused at 675 °C with sodium peroxide and sodium hydroxide. The fused samples were then dissolved with nitric acid. The resulting liquids were diluted to 100 mL with water and analyzed for ICP-ES and alpha counting. Each digestion was performed in quadruplicate. See Procedure L16.1-ADS-2226[5] for aqua regia and L16.1-ADS-2502[6] for peroxide fusion in the SRNL-ATFC procedure manual for detail of the digestion techniques.

For the elements Ag, B, Be, Cd, Ce, Co, Cu, Li, Mo, P, S, Sb, Sn, Ti, V, and Zn, ICP-ES results from AR digested samples are reported since these elements were not detected in the PF digested sample (AR digested samples typically have lower detection limits due to less dilution required at the instrument). For the elements not detected in either the PF or AR digestions, the lowest detection limit was reported. For K and Na, results from the ICP-ES of the AR digestions were used because these elements were detected in significant quantities in the PF reagent blank. The PF digestion method utilizes sodium peroxide, which, in addition to Na, likely contains K. The PF digestion method is performed in Zr crucibles, which contributes additional Zr to the analysis. For Hg, results from the DMA analysis of the AR digestion were used. For Th, ICP-ES from the AR digestion was reported because the result agreed well with ICP-MS results (mass 232). For Al, Ca, and Si, ICP-ES from the PF digestions were used because AR does not fully dissolve these elements.

Fission product elements Ru, Rh, and Pd were quantified by ICP-MS from the AR digestion using a methodology described by Bibler.[7] In summary, Ru is reported from the sum of masses 101, 102, and 104. Rh is taken from mass 103. Pd is determined from mass 105 and a factor derived from fission yields of other Pd isotopes.

Nd is also determined from ICP-MS from the AR digestion. Stable isotopes of Nd include masses 142 to 146, 148, and 150.[8] However, there is a stable isotope of Ce at mass 142. Therefore, mass 142 is not used in the Nd quantification, which may lead to small under-reporting of Nd.

Isotopes of uranium in the slurry were determined from U233/U234/U235/U236 through ICP-MS on PF digested samples. Isotopes of plutonium were determined from alpha counting and ICP-MS of PF digested samples.

For the elements quantified by ICP-ES, results from two digestions were compared. In general, if the percent difference between the two results was 10% or less, the average is reported. If the percent difference between results was greater than 10%, one digestion was chosen. Those decisions are noted in the results section.

2.3 Quality Assurance

Requirements for performing reviews of technical reports and the extent of review are established in manual E7 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2.[9] The functional classification requested for this work is Safety Class. In accordance with Manual E7, Procedure 3.60, a technical review of the

contents of this report has been performed to ensure consistency and accuracy. Per Manual E7, Procedure 2.60, a Design Verification, at a minimum, must be performed for work supporting a Safety Class functional classification.

3.0 Results and Discussion

3.1 Density and Weight % Solids

Presented in Table 3-1 are the first set of density and weight percent solid results.

Table 3-1 Densities and Weight Percent Solids

	Average	RSD, n*
Slurry Density (g/mL)	1.275	0.46, 3
Filtrate Density (g/mL)	1.214	0.20, 3
Wt% Total Solids (Slurry Basis)	31.5	0.46, 4
Wt% Dissolved Solids (Supernate Basis)	26.6	0.31, 4
Wt% Insoluble Solids (Slurry Basis)	6.7	N/A
Wt% Soluble Solids (Slurry Basis)	24.9	N/A
Wt% Calcined Solids (Slurry Basis)	22.0	4.0, 3

*RSD=relative standard deviation with number of measurements (n). NA= not applicable because result is calculated.

Weight % solids and densities performed on the combined sample revealed that the insoluble solids concentration was lower than SRMC projections. Due to this, additional weight % solids and density measurements were performed. Table 3-2 shows the additional densities and weight % solids.

Table 3-2 Additional Densities and Weight Percent Solids

	Average	RSD, n*
Slurry Density (g/mL)	1.288	0.12, 3
Filtrate Density (g/mL)	1.211	0.03, 3
Wt% Total Solids (Slurry Basis)	32.0	0.67, 4
Wt% Dissolved Solids (Supernate Basis)	26.6	0.29, 4
Wt% Insoluble Solids (Slurry Basis)	7.3	N/A
Wt% Soluble Solids (Slurry Basis)	24.7	N/A

*RSD=relative standard deviation with number of measurements (n). NA= not applicable because result is calculated.

Error propagation of the total solids (slurry basis) and soluble solids (filtrate basis) yielded a 95% confidence interval for the insoluble solids of approximately ±0.3 weight % for both sets performed. The confidence interval is derived using the inverse Student's t-distribution which requires probability and degrees of freedom, partial derivative of the weight percent of insoluble solids with respect to total solids, partial derivative of the weight percent of insoluble solids with respect to soluble solids, standard error of the weight percent total solids, and standard error of the weight percent soluble solids. Uncertainty for the insoluble solids does not account for variability in the sampling. Contributing factors to larger 95% confidence interval values include standard deviation of results (incorporated in standard error), number of replicates (incorporated in degrees of freedom), and average weight percent of soluble solids (incorporated in partial derivative of weight percent insoluble solids with respect to

total solids and soluble solids). A higher standard deviation of results, larger amount of weight percent soluble solids, and fewer number of replicates can lead to a larger 95% confidence interval value.

3.2 Settling Test

Table 3-3 shows the values collected of the slurry and supernate levels. These values of slurry height vs. time were graphed in Figure 3-1. Figure 3-2 depicts images of the settling test that lasted for 12 days and 17 hours.

Figure 3-1 HTF-24-51-85/86 Settling Test: Slurry Height vs. Time

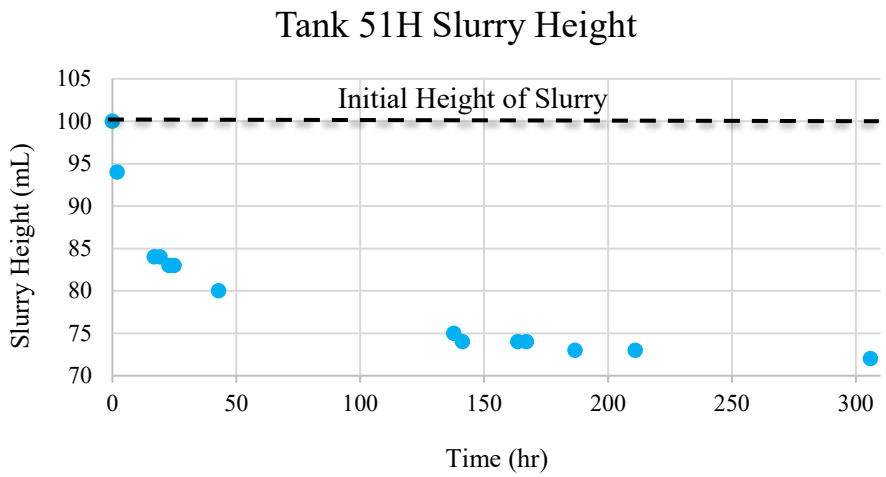
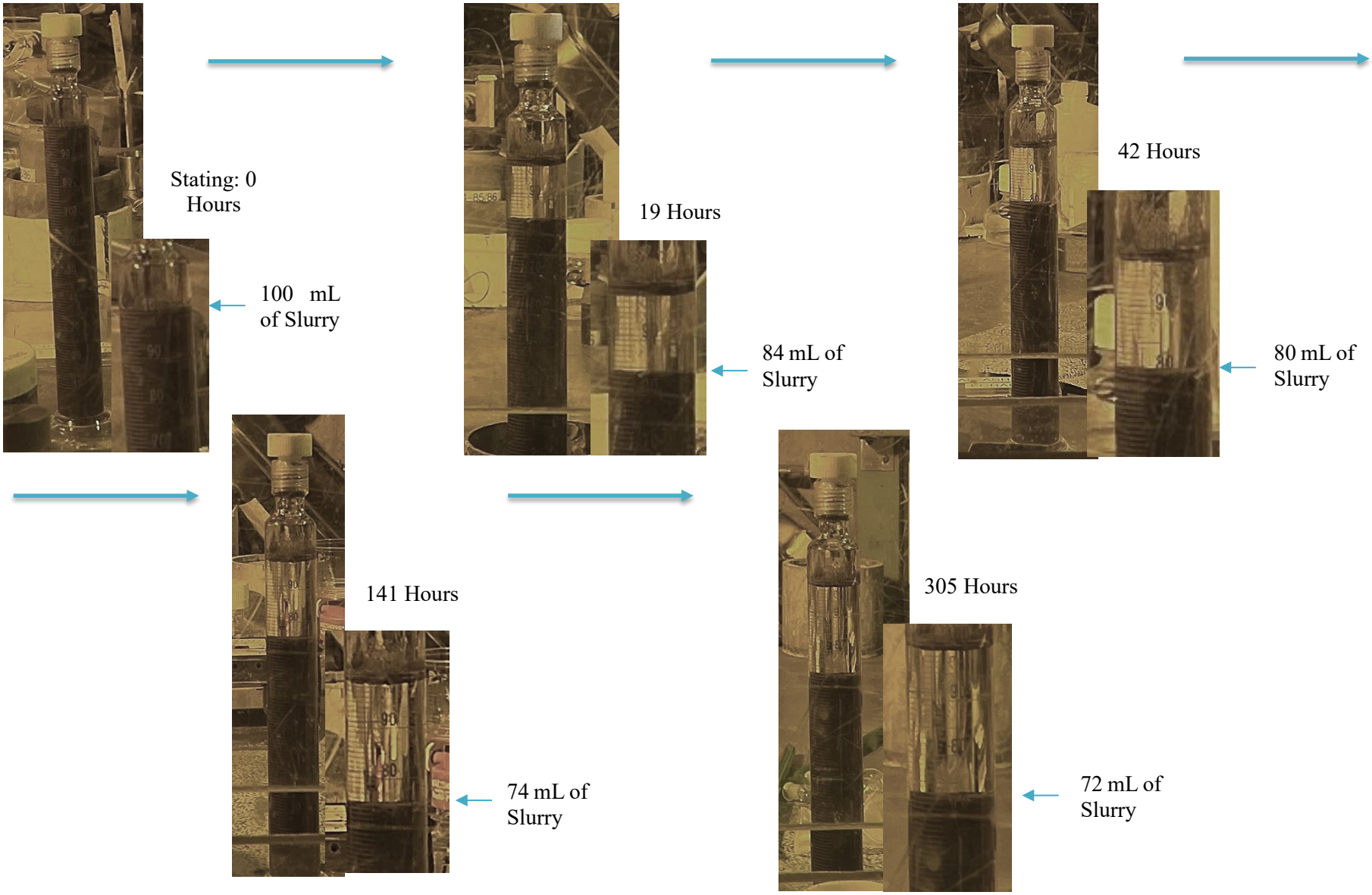


Table 3-3 HTF-51-24-85/86 Settling Test Table of Slurry and Supernate Levels

Hours	Slurry Level (mL)	Supernate Level (mL)
0	100	0
2	94	6
16.98	84	16
19.28	84	16
22.90	83	17
24.9	83	17
42.88	80	20
137.77	75	25
141.33	74	26
163.57	74	26
167.15	74	26
186.75	73	27
211.00	73	27
305.90	72	28

Figure 3-2 Tank 51H Slurry Height



3.3 Supernate Analytical Results

Table 3-4 shows the results for Supernate Anions and Cs-137, Table 3-5 shows the results for Elemental Composition of Supernate and Table 3-6 shows the results for Isotopes of Uranium and Plutonium of Supernate.

Table 3-4 Supernate Anions and Cs-137 Results

Analyte	Analytical Method	Units	Results	%RSD	One sigma uncertainty (%)
Bromide	IC	mg/L	<50	N/A	N/A
Chloride	IC	mg/L	1.44E+02	0.97	10
Fluoride	IC	mg/L	1.05E+03	0.74	10
Formate	IC	mg/L	<10	N/A	N/A
Nitrate	IC	mg/L	8.99E+04	4.84	10
Nitrite	IC	mg/L	2.58E+04	1.01	10
Oxalate	IC	mg/L	7.00E+02	0.71	10
Phosphate	IC	mg/L	<10	N/A	N/A
Sulfate	IC	mg/L	1.15E+04	0.59	10
Carbonate	TIC	mg/L	2.23E+04	5.01	7
Free OH	Titration	M	1.54E+00	3.06	5
Total Organic Carbon	TOC	mg Carbon/L	4.46E+02	3.86	7
Cs-137	Gamma-scan	dpm/mL	3.73E+08	1.55	5

*RSD=relative standard deviation with number of measurements (n). n=4 NA= not applicable

Table 3-5 Elemental Composition of Supernate

Element	Analytical Method	Results (mg/L)	% RSD, n*	One Sigma Uncertainty (%)
Ag	ES	<2.03E-01	N/A	N/A
Al	ES	5.32E+03	0.38, 4	10
B	ES	5.26E+01	0.18, 4	10
Ba	ES	<2.03E-01	N/A	N/A
Be	ES	<9.13E-02	N/A	N/A
Ca	ES	<5.83E-02	N/A	N/A
Cd	ES	<8.65E-02	N/A	N/A
Ce	ES	<2.20E+00	N/A	N/A
Co	ES	<4.05E-01	N/A	N/A
Cr	ES	2.35E+02	0.43, 8	10
Cu	ES	<8.79E-01	N/A	N/A
Fe	ES	<6.81E-01	N/A	N/A
Gd	ES	<2.04E-01	N/A	N/A
Hg	DMA	1.48E+02	9.86, 4	10
K	ES	6.37E+02	0.39, 4	10
La	ES	<3.45E-01	N/A	N/A
Li	ES	<9.85E+00	N/A	N/A
Mg	ES	<1.15E+00	N/A	N/A
Mn	ES	<3.30E-01	N/A	N/A
Mo	ES	2.51E+01	0.40, 4	10
Na	ES	1.10E+05	0.20, 4	10
Nd	MS	1.49E-01	N/A	N/A
Ni	ES	<6.11E-01	N/A	N/A
P	ES	8.21E+01	0.31, 4	10
Pb	MS	3.99E-01	3.50, 4	N/A
Rh	MS	4.62E-01	0.27, 4	10
Ru	MS	1.10E+00	1.01, 4	10
S	ES	4.00E+03	0.29, 4	10
Sb	ES	<3.03E+00	N/A	N/A
Si	ES	<2.07E+01	N/A	N/A
Sn	ES	<8.56E+00	N/A	N/A
Sr	ES	<9.77E-02	N/A	N/A
Th	ES	<3.81E+00	N/A	N/A
Ti	ES	<6.25E-01	N/A	N/A
U	ES	<1.08E+01	N/A	N/A
V	ES	<4.88E+00	N/A	N/A
Zn	ES	<1.42E+00	N/A	N/A
Zr	ES	<1.30E-01	N/A	N/A

*RSD=relative standard deviation with number of measurements (n). NA= not applicable

Table 3-6 Isotopes of Uranium and Plutonium of Supernate

Isotope	Results (dpm/mL)	%RSD, n*	One Sigma Uncertainty (%)
U-233	<1.20E+01	2.10	10
U-234	1.55E+01	10.16	10
U-235	1.82E-01	21.87	10
U-236	4.36E-01	24.90	10
U-238	6.64E+00	1.75	10
Pu-238	6.94E+03	10.47	14.03
Pu-239	<7.05E+02	N/A	N/A
Pu-240	<7.05E+02	N/A	N/A
Pu-241	<2.11E+04	N/A	N/A

*RSD = relative standard deviation with number of measurements (n). n = 4. N/A = not applicable

3.4 Slurry Analytical Results

Slurry oxalate results are reported in Table 3-7. Elements quantified by ICP-ES, ICP-MS, and DMA for slurry elemental composition are reported in Table 3-8. Total solids from Table 3-1 were used to calculate out total solids (wt%) in Table 3-8 and Table 3-9. Specific digestions and analytical techniques for each element are given with the results. Isotopes of uranium and plutonium are reported in Table 3-9.

Table 3-7 Slurry Oxalate Result

Analyte	Results (mg/L)	% RSD, n*	One Sigma Uncertainty (%)
Oxalate	4.98E+03	1.15	10

*RSD=relative standard deviation with number of measurements (n). n = 4

Table 3-8 Slurry Elemental Composition

Element	Digestion and Analytical Method	Results (µg/g of Slurry)	Total Solids (wt%)	%RSD, n*	One Sigma Uncertainty (%)
Ag	AR, ES	<2.05E+01	<6.50E-03	N/A	N/A
Al	PF, ES	1.55E+04	4.93E+00	0.32, 4	10
B	AR, ES	4.04E+01	1.28E-02	2.26, 4	10
Ba	AR/PF, ES	9.39E+01	2.98E-02	0.60, 8	10
Be	AR, ES	<2.67E-01	<8.47E-05	N/A	N/A
Ca	PF, ES	1.49E+03	4.73E-01	1.45, 4	10
Cd	AR, ES	1.29E+01	4.08E-03	1.60, 4	10
Ce	AR, ES	<1.60E+02	<5.08E-02	N/A	N/A
Co	AR, ES	<7.57E+00	<2.40E-03	N/A	N/A
Cr	AR/PF, ES	4.54E+02	1.44E-01	1.48, 8	10
Cu	AR, ES	<8.71E+01	<2.76E-02	N/A	N/A
Fe	AR/PF, ES	1.66E+04	5.25E+00	1.28, 8	10
Gd	AR/PF, ES	3.59E+02	1.14E-01	2.08, 8	10
Hg	AR, DMA	4.71E+03	1.49E+00	2.29, 4	10
K	AR, ES	4.49E+02	1.42E-01	1.83, 4	10
La	AR/PF, ES	5.85E+01	1.85E-02	2.90, 8	10
Li	AR, ES	<6.68E+01	<2.12E-02	N/A	N/A
Mg	AR/PF, ES	5.08E+02	1.61E-01	1.74, 8	10
Mn	AR/PF, ES	3.01E+03	9.56E-01	2.35, 8	10
Mo	AR, ES	2.25E+01	7.14E-03	4.25, 4	10
Na	AR, ES	8.21E+04	2.60E+01	1.76, 4	10
Nd	AR, MS	2.16E+02	6.85E-02	7.25, 4	N/A
Ni	AR/PF, ES	5.46E+02	1.73E-01	1.56, 8	10
P	AR, ES	<2.44E+02	<7.74E-02	N/A	N/A
Pb	AR, MS	6.44E+00	2.04E-03	7.51, 4	10
Rh	AR, MS	1.46E+01	4.64E-03	7.64, 4	10
Ru	AR, MS	7.92E+01	2.51E-02	7.36, 4	N/A
S	AR, ES	3.03E+03	9.62E-01	1.69, 4	10
Sb	AR, ES	<6.03E+00	<1.91E-03	N/A	N/A
Si	PF, ES	1.19E+03	3.77E-01	1.26, 4	10
Sn	AR, ES	<1.53E+01	<4.85E-03	N/A	N/A
Sr	AR/PF, ES	3.81E+01	1.21E-02	3.90, 4	10
Th	AR, ES	8.44E+02	2.68E-01	1.50, 4	10
Ti	AR, ES	3.85E+01	1.22E-02	2.22, 4	10
U	AR/PF, ES	6.01E+03	1.91E+00	6.70, 4	10
V	AR, ES	<9.70E+00	<3.08E-03	N/A	N/A
Zn	AR, ES	3.21E+01	1.02E-02	4.35, 4	10
Zr	AR, ES	1.54E+02	4.87E-02	15.44, 4	10

*RSD=relative standard deviation with number of measurements (n). NA= not applicable

Table 3-9 Isotopes of Uranium and Plutonium of Slurry

Isotope	Results (dpm/g of Slurry)	Total Solids (wt%)	%RSD, n*	One Sigma Uncertainty (%)
U-233	4.17E+03	6.16E-05	0.36	10
U-234	4.07E+04	9.31E-04	1.86	10
U-235	5.51E+02	3.65E-02	0.62	10
U-236	1.71E+03	3.78E-03	0.56	10
U-238	4.70E+03	2.00E+00	0.56	10
Pu-238	1.14E+08	9.52E-04	4.27	4.74
Pu-239	2.92E+06	6.72E-03	5.33	13.4
Pu-240	1.43E+06	8.97E-04	6.27	19.5
Pu-241	3.33E+07	4.62E-05	4.81	14.9

*RSD = relative standard deviation with number of measurements (n). n = 4.

3.5 Total and Soluble Na, Cr, Hg and S

Total Cr, Na, and S were obtained from AR digestions of the slurry with ICP-ES. Total Hg was obtained from AR digestions of the slurry with DMA. Soluble Cr, Hg, Na, and S were measured from the supernate elemental results in Table 3-4. Supernate basis was corrected to slurry basis with supernate density and wt% insoluble solids from Table 3-1.

Table 3-10 Soluble Sodium, Chromium, Sulfur, and Mercury in HTF-51-24-85/86

Element	Total (µg/g Slurry)	Soluble (mg/kg Slurry)	% Soluble*
Cr	4.54E+02	1.80E+02	4.0E+01
Hg	4.71E+03	1.13E+02	2.4E+00
Na	8.21E+04	8.49E+04	1.0E+02
S	3.03E+03	3.08E+03	1.0E+02

*% Soluble values were rounded to 2 significant fiqure due to uncertainty in the individual measurements.

4.0 Conclusions

SRNL has been requested to qualify the next sludge batch, SB11 for processing at DWPF. SRMC sent SRNL two 200 mL samples received from Tank 51H on the 12th of August 2024. This report contains analytical results for the as-received combined qualification sample, HTF-51-24-85/86. The sample was analyzed for: supernate and slurry density, weight percent solids, settling, chemical composition, and radionuclides. Weight % solids and densities performed revealed that the insoluble solids concentration was lower than SRMC projections. Due to this, additional weight % solids and density measurements were performed. This resulted in higher insoluble solids and lined up with SRMC's projection. All other results from chemical composition and radionuclides were comparable to SRMC's projections.

5.0 References

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Distribution:

John.pareizs@srnl.doe.gov
Anthony.howe@srnl.doe.gov
Dan.lambert@srnl.doe.gov
Matthew.siegfried@srnl.doe.gov
Matthew.williams@srnl.doe.gov
Wesley.woodham@srnl.doe.gov
Seth.hunter@srnl.doe.gov
Fabienne.johnson@srnl.doe.gov
Alex.cozzi@srnl.doe.gov
Sean.noble@srnl.doe.gov
Elizabeth.craig@srnl.doe.gov
William.bates@srnl.doe.gov
Brandi.clark@srnl.doe.gov
c.diprete@srnl.doe.gov
David.diprete@srnl.doe.gov
connie.herman@srnl.doe.gov
sarah.hodges@srnl.doe.gov
brady.lee@srnl.doe.gov
kalyn.loudermilk@srnl.doe.gov
Heather.Capogreco@srnl.doe.gov
Joseph.Manna@srnl.doe.gov
Gregg.Morgan@srnl.doe.gov
frank.pennebaker@srnl.doe.gov
Amy.Ramsey@srnl.doe.gov
William.Ramsey@srnl.doe.gov
eric.skidmore@srnl.doe.gov
michael.stone@srnl.doe.gov
boyd.Wiedenman@srnl.doe.gov
Mary.whitehead@srnl.doe.gov
Jennifer.Wohlwend@srnl.doe.gov
chris.martino@srnl.doe.gov
crystal.currie@srnl.doe.gov
Records Administration (EDWS)
terri.fellinger@srs.gov
robert.hoeppel@srs.gov
bill.holtzscheiter@srs.gov
Vijay.Jain@srs.gov
Jeremiah.Ledbetter@srs.gov
jeff.ray@srs.gov
aubrey.silker@srs.gov
spencer.isom@srs.gov
Melany.mini@srs.gov
phoebe.fogelman@srs.gov
David.henley@srs.gov